Local Floer Chain Complex across Hamiltonian Bifurcation

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Contents

This talk is based on the ongoing work with Hong-kwon Jo (SNU).

- **Motivation and Preliminaries.**
  Bifurcation, Conley–Zehnder index and Floer homology.
- **Classification of Hamiltonian bifurcations.**
  Birth-death, period doubling, phantom kiss and emission.
- **Chain complex mutation.**
  Construction of cylinders, Morse–Bott cascade counting.

Main result. Describing the explicit change of Conley–Zehnder indices and Floer chain complex for generic Hamiltonian bifurcations.

Motivation

Motivation 1. Bifurcations

A **bifurcation** is, heuristically, a birth and death of periodic orbits.

Recently, there are many works [AFvK^{+}23], [AB25], [JKvK26] relating bifurcation with symplectic invariants, mostly Conley–Zehnder indices.

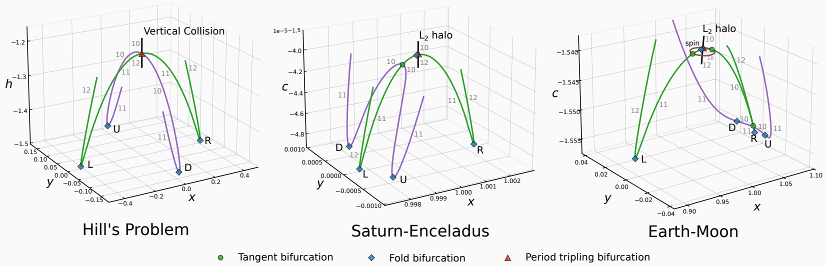


Figure 1: Bifurcation in Hill’s lunar problem and CRTBP, from [JKvK26].

Motivation 2. Rotating Kepler Problem

In the rotating Kepler problem, there are three important periodic orbits: retrograde, direct and vertical collision orbits.

The multiple covers of the retrograde and direct orbits bifurcates, which produces orbits looks like a flower.

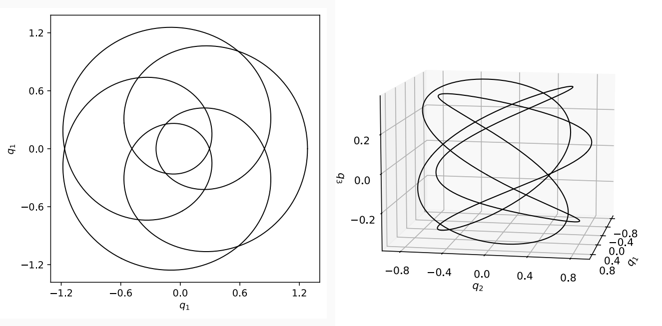
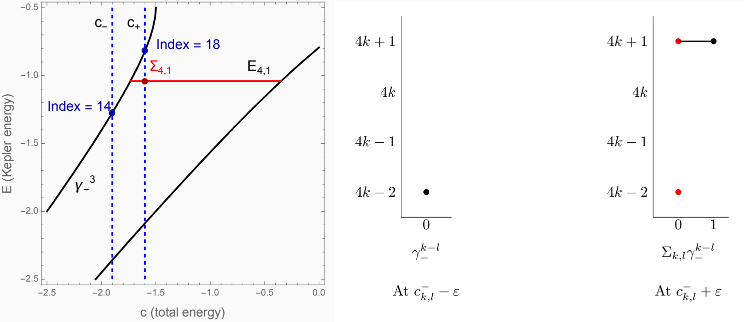


Figure 2: Resonant orbits of RKP, in courtesy of Chankyu Joung.

Motivation 2. Rotating Kepler Problem

In my previous work [Lee26], I observed this bifurcation can be translated to a change of Floer chain complex in terms of spectral sequences.



Slogan. Bifurcation gives a *mutation of a Floer chain complex*.

Preliminaries

Toy Example. Critical Points of a Function

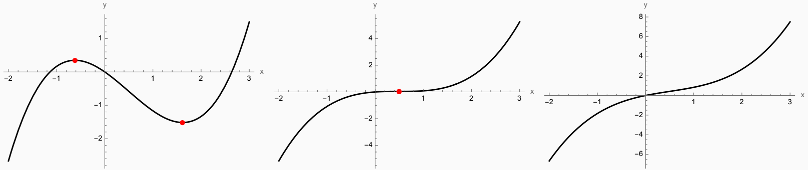


Figure 3: Birth-death bifurcation of critical points

- Consider a one-parameter family of functions, $f_\varepsilon(x) = x^3 + \varepsilon x$.
- As ε passes through 0, a minimum and a maximum meet and disappear. This is called the **birth-death** of critical points.
- The same kind of phenomenon happens for **periodic orbits** of Hamiltonian systems.

Hamiltonian Periodic Orbits

Let $H : W^4 \to \mathbb{R}$ be a Hamiltonian and γ be a periodic orbit.

Poincaré section S of γ is a transversal section of γ in $H^{-1}(c)$.

The system is locally simplified to a **return map** Θ on S.

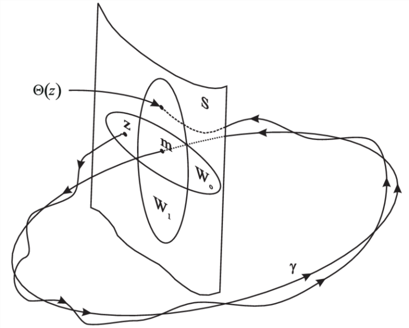


Figure 4: Poincaré section and the return map, from [OR99].

Floquet Multipliers

Floquet multiplier of γ is an eigenvalue of the linearized return map.

The map is symplectic, so we have a quartet $\{\lambda, \bar{\lambda}, \lambda^{-1}, \bar{\lambda}^{-1}\}$, and actually only 2 possibilities since the return map is 2-dimensional.

- **Elliptic orbit**: $\lambda \in S^1$, or $\lambda^{-1} = \bar{\lambda}$.
- **Hyperbolic orbit**: $\lambda \in \mathbb{R} \setminus \{0\}$, or $\lambda = \bar{\lambda}$.

If $\lambda \neq 1$, we call γ is **non-degenerate**.

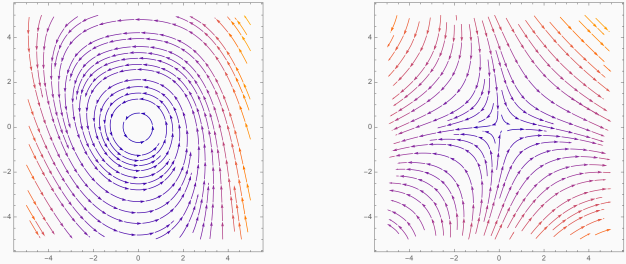


Figure 5: Elliptic and hyperbolic fixed points

Bifurcation

Let $H_\varepsilon : W \to \mathbb{R}$ be a 1-parameter family of Hamiltonians, and let γ_0 be a periodic orbit of H_0 with Floquet multiplier λ_0.

If $\lambda_0 \neq 1$, then nearby periodic orbits form an orbit cylinder

$$\Gamma = \{(\gamma_\varepsilon, \varepsilon) : \gamma_\varepsilon \text{ is a periodic orbit of } H_\varepsilon\} \simeq S^1 \times I.$$

This is a *continuation* of the orbit along bifurcation parameter.

If the orbit is degenerate, the orbit cylinder may break:

- the stability type may change,
- periodic orbits may be born or vanish, called **bifurcation**,
- if $\lambda \neq 1$ but $\lambda^k = 1$, then the k-th cover bifurcates.

Conley–Zehnder Index

The **Conley–Zehnder index** $\mu_{\mathrm{CZ}}(\gamma)$ is an integer associated to a periodic orbit, might be understood as a *symplectic winding number*.

Along a non-degenerate family, the CZ index is constant.

$\Rightarrow$ CZ index changes only when the orbit becomes degenerate, and it is exactly when the bifurcation happens.

Idea. We can analyze bifurcations via detecting CZ indices.

Floer Homology

A **Floer homology** $HF(W, H)$ is a topological invariant associated to a symplectic manifold W and Hamiltonian H.

This is determined by **Floer chain complex** $CF(W, H)$ whose

- generators are 1-periodic orbits of H,
- grading is given by CZ indices,
- differential is determined by a count of **Floer cylinder**.

Across a bifurcation, $HF(W, H)$ doesn't change but $CF(W, H)$ does.

- number of generators changes (orbits are born or vanish)
- grading changes (change of CZ index)

Hence, the differential should be re-arranged the chain complex to determine the same homology. *Can we find this explicitly?*

Classification of Bifurcations

Meyer's Classification

Generic Hamiltonian bifurcations are classified in [Mey70].

They are organized by the covering number k of the orbit.

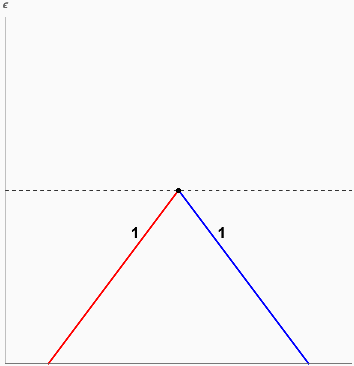


Figure 6: Birth-death, $k = 1$

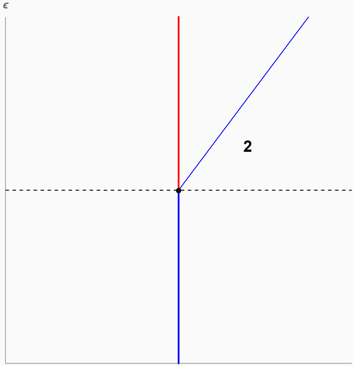


Figure 7: Period doubling, $k = 2$

Hyperbolic is red, elliptic is blue and numbers are approximate periods.

Meyer's Classification

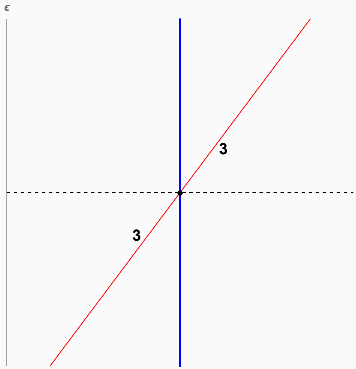


Figure 8: Phantom kiss, $k = 3, 4$

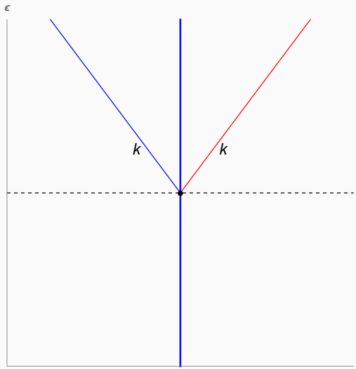


Figure 9: Emission, $k \geq 4$

Note. Period doubling with the opposite stability is called **murder**.

Method of Classification

We find the periodic points of the return map by **Birkhoff normal form**.

This can also be done in essentially same way, by finding generating Hamiltonian of the return maps and finding the critical point.

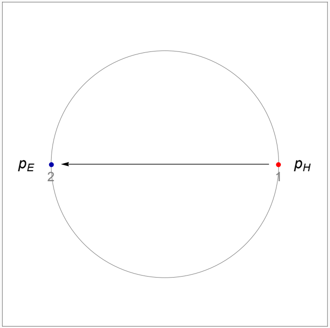


Figure 10: Birth-death

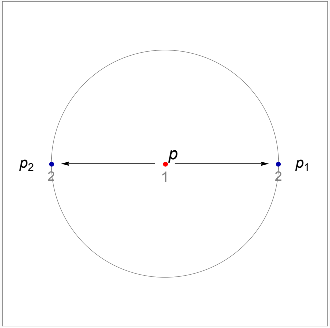


Figure 11: Period doubling

The figure indicates critical points and gradient flows on Poincaré section.

Method of Classification

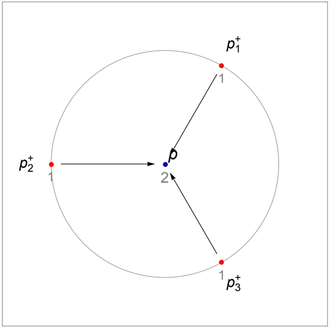


Figure 12: 3-phantom kiss

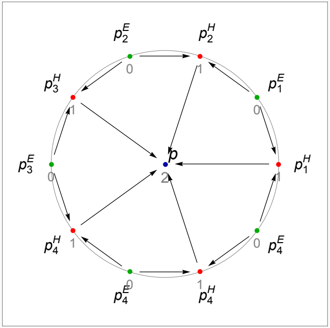


Figure 13: 5-emission

Same colored k-points indicates one k-periodic orbit.

Chain Complex Mutation

Easy example. Birth-death bifurcation

Let an orbit γ_0 experiences the birth-death bifurcation along ε.

1. If $\varepsilon < 0$, there is no orbit.
2. If $\varepsilon = 0$, they degenerate and merge into a single orbit γ_0.
3. If $\varepsilon > 0$, there are two orbits, say γ_H and γ_E.

(Very simplified) **Floer-theoretic viewpoint.**

1. If $\varepsilon < 0$, $CF = 0$.
2. If $\varepsilon > 0$, $CF = \langle \gamma_H, \gamma_E \rangle$. But both produces same homology.

Hence, $|\mu_{CZ}(\gamma_H) - \mu_{CZ}(\gamma_E)| = 1$ and they should cancel each other, so there exists a Floer cylinder between them. (cf. [GG10].)

However, the situations are more complicated for higher covers.

Adiabatic approximation of Hamiltonian

Let $\tilde{H}_\varepsilon$ be a Hamiltonian, γ_0 be a periodic orbit and Θ_ε be return maps.

If $\lambda_0^k = 1$, we have an autonomous generating Hamiltonian G_ε of Θ_ε^k, produced by Birkhoff normal form.

Using **adiabatic approximation**, which is an integration along the orbit, we construct the model Hamiltonian for each bifurcation type, say H_ε.

This Hamiltonian is not exactly same as the original one, but has the same return map and Conley–Zehnder indices with $\tilde{H}_\varepsilon$.

Note. The Floer chain complex is not fully determined by W and H; it also depends on a choice of other auxillary functions.
Thus, we constructed a representative model describes a bifurcation.

Morse–Bott Cascade

Our periodic orbits have S^1-symmetry by time shift.
(To avoid this, Floer theory uses time-dependent Hamiltonian.)

Instead, we equip an auxillary perturbation on each orbit, whose critical points correspond to periodic orbits.

In this case, differential is given by counting **Morse–Bott cascade**, a sequence of Morse trajectories and Floer cylinders.

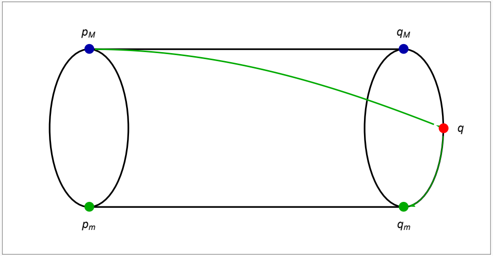


Figure 14: Morse–Bott cascade.

Key Ideas

- Gradient trajectories of G_ε gives Floer cylinders of H_ε.

We get cylinders connecting a simple orbit and a multiple cover.

$\Rightarrow$ A contribution of 1 cylinder may not be exactly 1.

Using Morse–Bott cascade, we can compute the explicit chain complex, of which the resulting homology is invariant.

Example. Period doubling

1. If $\varepsilon < 0$, there is one orbit γ.
2. γ^2 bifurcates at $\varepsilon = 0$ and index changes, say 1 to 0.
3. If $\varepsilon > 0$, there are two orbits; γ^2 and γ_H.
4. The index of γ_H must be 0, and there exists 1 Floer cylinder between γ^2 and γ_H, whose contribution is given as following.

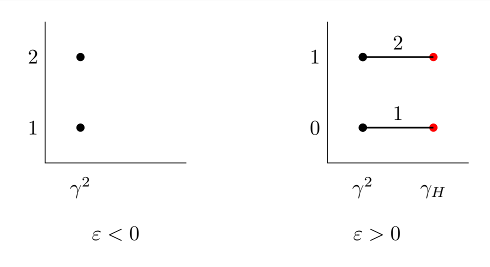


Figure 15: Chain complex mutation across the period doubling.

$\gamma_{H,\min}$ and $\gamma^2_{\min}$ cancel each other, while $\gamma_{H,\max}$ and $\gamma^2_{\max}$ don't.

Possible applications

Dynamical direction.

- Making profile of index changes for generic bifurcation types.
- Applying the framework to symmetric bifurcations, such as pitchfork.
- Applying for multi-variable or higher-dimensional bifurcations.

Floer direction.

- Analyzing the change of moduli space of Floer cylinders across bifurcations. (We need more global arguments here.)

Thank you!

Questions?

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